26. Define:
(i) Event, (ii) Random Experiment, (iii) Sample space, (iv) Mutually exclusive events, (v) Exhaustive events (vi) Independent events, (vii) Dependent events.

27. Define mathematical probability.

28. Define statistical probability.

29. Define conditional probability.

30. State the Multiplication theorem on probability for any two events.

31. State the Multiplication theorem of probability for independent events.

IV. Short Answer Questions:

32. What is the chance that a non-leap year selected at random will contain 53 Sundays or 53 Mondays?

33. When two dice are thrown, find the probability of getting doublets. (same number on both dice).

34. A box containing 5 green balls land 3 red colour balls. Find the probability of selecting 3 green colour balls without replacement.

35. There are 5 items defective in a sample of 30 items. Find the probability that an item chosen at random from the sample is (i) defective (ii) non – defective

36. Given that $P(A) = 0.35$, $P(B) = 0.73$ and $P(A \cap B) = 0.14$, find $P(\overline{A} \cup \overline{B})$

37. State the axioms of probability.

38. State the theorem on total probability.

39. State Bayes' theorem.

40. A card is drawn at random from a well shuffled pack of 52 cards. What is the probability that it is (i) an ace (ii) a diamond card?

V Calculate the following:

41. State and prove addition theorem on probability.

42. An urn contains 5 red and 7 green balls. Another urn contains 6 red and 9 green balls. If a ball is drawn from any one of the two urns, find the probability that the ball drawn is green.